

ADVANCE PROGRAMMING

JUNIT TUTORIAL

In this tutorial, we will learn how to use the JUnit 5 feature to test our code. It is not an inbuilt library in Java, and *would have to be installed separately*. However, if you use IntelliJ or Eclipse, then the process of integrating JUnit with your project might become simpler. This tutorial does not focus on the installation part, as the focus would be to use its features to test the code.

Question. You have to do the following:

- 1) Create a class with two methods: square, and countA.
- 2) square takes an integer argument, and returns the square of the number.
- 3) countA takes a String as argument, counts the number of A's in it (both uppercase and lowercase), and returns this count.
- 4) Now you have to create 2 more classes: one to test the square method, and the other to test the countA method.
- 5) For the square() method, take 3, -4 and 11 as your test inputs, and assert them to be equal to 12, 16, and 121 respectively. How many test cases pass?
- 6) For the countA() method, take "Anchorage", "ALABAMA" and "aajfihah" as your test inputs, and assert them to be equal to 11, 4, and 9 respectively. How many test cases pass?

NOTE: As an extra exercise, explore how you can combine these test cases into a test suite.

SOLUTION:

```
import java.beans.Transient;

import org.junit.Assert.*;

public class JUnitTutorial{
    public int square(int num){
        return num*num;
    }
    public int countA(String str){
        int len = str.length();
        int count = 0;
        for(int i=0;i<len;i++){
```

```

        if(str.charAt(i)=='a' || str.charAt(i)=='A'){
            count++;
        }
    }
    return count;
}

public class squareTest{
    @Test
    public void test(){
        JUnitTutorial test = new JUnitTutorial();
        int opt1 = test.square(3);
        int opt2 = test.square(-4);
        int opt3 = test.square(11);
        assertEquals(12, opt1);
        assertEquals(16, opt2);
        assertEquals(121, opt3);
    }
}

public class countATest{
    @Test
    public void test(){
        JUnitTutorial test = new JUnitTutorial();
        int opt1 = test.countA("Anchorage");
        int opt2 = test.countA("ALABAMA");
        int opt3 = test.countA("aajfihah");
        assertEquals(11, opt1);
        assertEquals(4, opt2);
        assertEquals(9, opt3);
    }
}

```

Note: These classes should be kept in separate files since they are all public.